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algebra-based physics-1: midterm 2

unit 4:

1.) B: 45 rpm, 4.7 radians sec^{-1}

2.) D: 2.2 $m\,s^{-2}$

3.) D: $\omega \rightarrow 10\,\omega$

4.) B: 0.53 N

5.) C: 42 $m\,s^{-1}$

unit 5:

1.) D: 200 N m, 37 degrees

2.) a. $W = F \cdot d$

$W = 1000\,N \cdot 0.1\,m$

$\boxed{W = 100\,J}$

b. $\boxed{\Delta K = 100\,J}$

c. $\Delta K = \frac{1}{2} m v_f^2 \rightarrow V_f = \sqrt{\frac{2\Delta K}{m}}$

$V_f = \sqrt{\frac{2 \times 100}{0.15}}$

$\boxed{V_f = 36.5\,m/s}$

d. $R = \frac{V_0^2 \cancel{\sin(2\theta)}^{\,1}}{g}$

$R = \frac{(36.5)^2}{9.81}$

$\boxed{R = 135.9\,m}$

3.) a. $f_k = \mu_k \cdot N$ $N = 40 \times 9.81 = 392.4N$ $f_k = .05 \times 392.4 = 19.62N$

$W = f_k \cdot d$

$W = 19.62 \times 40$

$\boxed{W = 784.8J}$

b. $P = \frac{W}{t}$

$= \frac{784.8}{30s}$

[redacted]

$\boxed{P = 26.16\ J/S}$

4.) D. $2.33

unit 6:

1.) C. 0 m/s

2.) $V_f = \frac{P_T}{m_1 + m_2} = \frac{0}{m_1 + m_2}$

$\boxed{V_f = 0m/s}$

3.) a. impulse $= F \cdot \Delta t$

$= 4000 \times 0.2s$

$= \boxed{800\ N \cdot s}$

b. $V_f = \frac{800}{200} + 2.80$

$\boxed{V_f = 6.80\ m/s}$

4.) C: elastic

5.) a. $(30,000)(0.85) + (110,000)(0) = (30,000 + 110,000)V_f$

$V_f = \frac{25500}{140000}$

$\boxed{V_f = 0.182\, m/s}$

b. $KE = \frac{1}{2}mv^2$

$KE = \frac{1}{2}(30000)(0.850)^2$ $KE = \frac{1}{2}(140000)(0.182)^2$

$KE = 10837.5 J$ → before $KE = 2318.68$ → after

$10837.5 - 2318.68 = \boxed{8518.82 J}$

unit 7:

1.) A: remain motionless

2.) $W = m \cdot g = 20 \times 9.81 = 196.2 N = F_1 + F_2$

$F_1 = \frac{98.1}{2}$ $196.2 - 49.05 = $ ↓

$\boxed{F_1 = 49.05 N}$ $\boxed{F_2 = 147.15 N}$

unit 8:

1.) a. $I = \frac{1}{2}MR^2$

$I = \frac{1}{2}(1)(0.08)^2$

$\boxed{I = 0.0032\, kg \cdot m^2}$

2.) $\boxed{a.\ -2.4\hat{k}\, Nm}$

$\boxed{b.\ -12\hat{j}\, N}$

b. $\tau = I\alpha$

$\tau = (0.0032)(2)$

$\boxed{\tau = 0.0064\, N \cdot m}$